# Globally consistent Land Environmental Assessment Factors (LEAFs) for soil organic carbon, soil erosion, and terrestrial acidification in support of SBTN Land v2 Targets

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**Abstract**

Setting credible, science-based land targets requires companies to compare the environmental performance of their sourcing landscapes against ecoregion-specific reference conditions, yet site-level primary data are rarely available across global supply chains. We present *Land Environmental Assessment Factors* (LEAFs): a global, openly licensed dataset of default factors for three soil-quality indicators —soil organic carbon (SOC), soil erosion, and terrestrial acidification—together with a reproducible Python pipeline (`sbtn_leaf`) that generates them. SOC factors are produced by running the Rothamsted Carbon (RothC) model forward to 2030 on a harmonized stack of SoilGrids, satellite climate, and crop-specific yield and residue inputs; soil-erosion factors apply the Revised Universal Soil Loss Equation (RUSLE) to Global Soil Erosion Modelling (GloSEM) base layers with crop and management cover-factors; and terrestrial-acidification factors are derived from published fate and sensitivity factors normalized to $SO_2$-equivalents. Factors are resolved for 42 land-use classes under up to eight management combinations (irrigation, tillage, and residue management) and are delivered as native-resolution rasters and as area-weighted summaries for terrestrial ecoregions, countries, and subcountry units. Across ecoregions the 2030 SOC factors average 34.7 t C/ha, baseline soil-erosion rates average $48.8\,\mathrm{t\,ha^{-1}\,yr^{-1}}$ (median 15.3), and acidification factors span more than an order of magnitude between biogeographic realms. Adopting reduced tillage raises projected SOC by on the order of 1.6 t C/ha (≈5%) for a representative cereal. All code, documentation, and factor tables are released under a CC0 license so that organizations can apply the factors directly or regenerate customized factors from primary data.

## 1 Introduction

Land use is among the largest drivers of biodiversity loss, soil degradation, and greenhouse-gas emissions, and it is increasingly central to corporate sustainability commitments. The Science Based Targets Network (SBTN) Land methodology asks companies to set targets that move their sourcing landscapes toward a "safe operating space" consistent with planetary boundaries (**??**). Operationally, this requires two ingredients: an ecoregion-specific *threshold* that defines safe conditions for a given soil-quality indicator, and an *assessment factor* that estimates the indicator value associated with a company's actual land use and management. While thresholds are derived from reference ecosystems, the assessment factor must be estimated for every combination of commodity, geography, and management practice a company may operate—a quantity that is almost never measured directly at scale.

Life-cycle assessment (LCA) provides regionalized characterization factors for related impacts, but these are typically tuned to product-level accounting rather than to the ecoregional, practice-sensitive comparisons that target-setting demands. There is therefore a need for *default* factors that are (i) globally complete, (ii) spatially explicit at ecoregional granularity, (iii) responsive to land-management choices, and (iv) transparent and reproducible so that organizations with primary data can refine them.

This paper introduces Land Environmental Assessment Factors (LEAFs) to meet that need. We build on the global SOC modelling framework of **?** and **?**, extend it to soil erosion and terrestrial acidification, and package the entire workflow as an open-source Python library. Our contributions are: (1) a harmonized, global, ecoregion-resolved dataset of default factors for three soil-quality indicators under multiple management scenarios; (2) a reproducible, tested software implementation (`sbtn_leaf`) covering data harmonization, biogeochemical and empirical modelling, spatial aggregation, and export; and (3) an analysis of the resulting global patterns and of the sensitivity of SOC factors to conservation practices. The factors are designed to plug directly into the SBTN Land accounting framework (**?**).

# 2 Materials and Methods

## 2.1 Spatial framework and overview

All factors are computed on a common spatial framework built from the Unique Homogeneous Thermal–Humidity (UHTH) zonation of **?**, which partitions the globe following the FAO Global Agro-Ecological Zones (GAEZ) suitability framework (**?**). Every input layer is harmonized—reprojected and resampled—to this common grid before calculation. After per-pixel factors are produced, they are aggregated by area-weighted zonal statistics to three reporting units: terrestrial ecoregions (**?**), countries, and subcountry administrative units. Reporting at ecoregion level is the default because it best matches the resolution of SBTN thresholds; country averages are retained only as a last resort because of their low spatial representativeness. The overall workflow comprises four stages—data gathering and processing, harmonization, factor calculation, and zonal aggregation—applied consistently to each indicator.

## 2.2 Input datasets

Table **??** summarizes the primary inputs. Climate drivers are multi-year means to represent typical rather than single-year conditions: monthly precipitation from NASA GPM IMERG averaged over 2013–2023 (**?**), and monthly temperature from the GLDAS Catchment Land Surface Model (**?**). Soil properties (organic-carbon stock for 0–30 cm, and clay and sand fractions for 15–30 cm) are taken from SoilGrids (**?**). Yields combine FAOSTAT national statistics (**?**) with the spatially disaggregated SPAM 2020 product (**?**). Soil-erosion base layers come from the GloSEM platform (**???**), and acidification fate–sensitivity factors from **?**.

## 2.3 Soil organic carbon factors

**RothC model.** SOC factors are produced with the Rothamsted Carbon (RothC) model (**?**), which simulates the turnover of organic carbon in non-waterlogged topsoils, accounting for soil

**Table 1:** Primary input datasets used to generate the LEAFs.

| Variable | Source | Native resolution / notes |
|---|---|---|
| Monthly precipitation | NASA GPM_3IMERGM (?) | 0.1°, 2013–2023 mean |
| Monthly temperature | GLDAS CLSM10 M v2.1 (?) | 1.0° |
| SOC stock (0–30 cm) | SoilGrids (?) | |
| Clay fraction (15–30 cm) | SoilGrids (?) | |
| Sand fraction (15–30 cm) | SoilGrids (?) | reduced-tillage only |
| $R \cdot K_{st} \cdot LS$ base | GloSEM v1.1/1.3 (?) | 25 km |
| Crop yields | FAOSTAT (?) + SPAM 2020 (?) | rainfed/irrigated splits |
| Residue parameters | IPCC (??) | Table ?? |
| Acidification FF×SF | ? | 2°×2.5° |
| Land-use / UHTH zones | ? | FAO GAEZ v2 |

type, temperature, moisture, and plant cover. RothC partitions soil carbon into five pools: decomposable plant material (DPM), resistant plant material (RPM), microbial biomass (BIO), humified organic matter (HUM), and inert organic matter (IOM). The active pools decompose with first-order kinetics whose monthly rate is modulated by three dimensionless rate-modifying factors. The temperature factor is

$$a = \frac{47.91}{1 + \exp\left(\frac{106.06}{T + 18.27}\right)}, \tag{1}$$

where $T$ is mean monthly air temperature (°C); a moisture factor $b$ is derived from the accumulated topsoil moisture deficit, which depends on rainfall, evapotranspiration, and a clay-dependent maximum deficit; and a plant-cover factor $c$ takes the value 1.0 for bare soil and 0.6 when the soil is vegetated. A modified tillage rate-modifying factor, adapted from ? using soil sand content and SOC, is applied to represent conservation (reduced) tillage. Carbon enters the system as monthly plant-residue inputs (and, optionally, farmyard manure), partitioned between DPM and RPM by a ratio of 1.44 for agricultural crops and improved grassland, 0.67 for unimproved grassland and scrub, and 0.25 for woodland (?). The model is initialized from the SoilGrids SOC baseline and clay content and integrated forward month by month; the SOC stock in the final projection year (2030 for the published factors) is the SOC LEAF, expressed in t C/ha.

**Potential evapotranspiration.** Because routine open-pan evaporation measurements are unavailable globally, RothC's moisture deficit is driven by potential evapotranspiration (PET) estimated with the Thornthwaite equation (?). For months with $T > 0\,°\mathrm{C}$,

$$\mathrm{PET} = 1.6 \cdot \frac{L}{12} \cdot \frac{N}{30} \cdot \left(\frac{10T}{I}\right)^{a_\mathrm{T}}, \tag{2}$$

where $L$ is mean daily daylight hours, $N$ the number of days in the month, $I$ the annual heat index $I = \sum_i (T_i/5)^{1.514}$ over months with $T_i > 0$, and $a_\mathrm{T} = 6.75 \times 10^{-7} I^3 - 7.71 \times 10^{-5} I^2 + 1.792 \times 10^{-2} I + 0.49239$. Daylight duration $L$ is obtained from the solar hour angle $\omega_0$, with $\cos\omega_0 = -\tan\phi\,\tan\delta$, latitude $\phi$, and solar declination $\delta = 23.44° \sin\big(\frac{360°}{365}(n - 81)\big)$ evaluated at the middle day $n$ of each month. The location-based PET is converted to a crop-specific value by multiplying by a monthly crop coefficient $K_c$, built from the FAO-56 single-coefficient method (?) using stage values $K_{c,\mathrm{ini}}$, $K_{c,\mathrm{mid}}$, and $K_{c,\mathrm{late}}$ defined per crop and thermal zone (??). Figure ?? illustrates a representative $K_c$ curve.

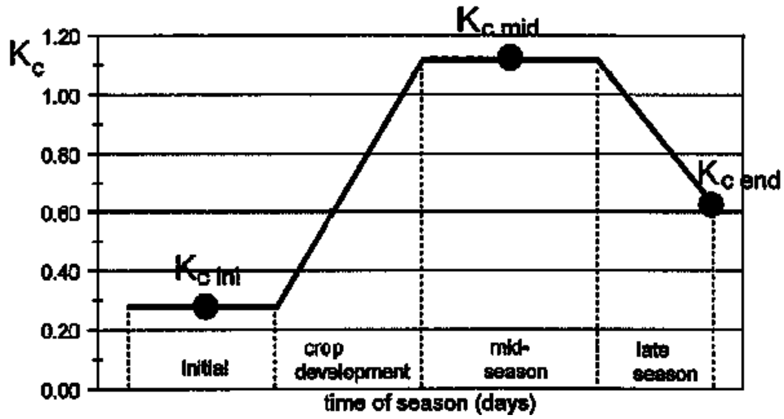


**Figure 1:** Example monthly crop-coefficient ($K_c$) curve constructed from FAO-56 stage values and stage durations (?), used to convert location-based potential evapotranspiration into crop evapotranspiration.

**Table 2:** IPCC residue parameters used to convert yields into above- and below-ground residue carbon inputs (??).

| Crop | Dry | Slope | Intercept | RS |
|---|---|---|---|---|
| Barley | 0.89 | 0.98 | 0.59 | 0.22 |
| Seed cotton (unginned) | 0.85 | 1.07 | 0.85 | 0.19 |
| Maize | 0.87 | 1.03 | 0.61 | 0.22 |
| Rape / colza seed | 0.85 | 1.13 | 0.85 | 0.19 |
| Sorghum | 0.89 | 0.88 | 1.33 | 0.22 |
| Soya bean | 0.91 | 0.93 | 1.35 | 0.19 |
| Sunflower seed | 0.85 | 1.13 | 0.85 | 0.19 |
| Wheat | 0.89 | 1.51 | 0.52 | 0.23 |

**Yields and residue inputs.** Plant-residue carbon inputs are derived from commodity yields. To assign a yield to every UHTH zone where a crop can grow, a hierarchical procedure is applied (implemented in `create_crop_yield_raster_withIrrigationPracticeScaling`): (1) national FAOSTAT yields and a calibration ratio to 2020 are computed; (2) SPAM 2020 rainfed, irrigated, and total yields are averaged and rescaled by the FAOSTAT ratio; (3) where SPAM data exist they are assigned, otherwise the FAOSTAT national value adjusted for irrigation practice is used; (4) remaining gaps are filled by ecoregion and then biome means; and (5) any residual gaps are filled by nearest-neighbour assignment. Yields are converted to dry matter and then to above- and below-ground residue carbon following the IPCC method (??):

$$\mathrm{AG}(T) = \mathrm{Crop}(T) \cdot \mathrm{Slope}(T) + \mathrm{Intercept}(T), \tag{3}$$

$$\mathrm{BG}(T) = \left[\mathrm{Crop}(T) + \mathrm{AG}(T)\right] \cdot \mathrm{RS}(T), \tag{4}$$

where $\mathrm{Crop}(T) = \mathrm{Yield}(T) \cdot \mathrm{DRY}(T)$, and the slope, intercept, dry-matter, and root-to-shoot (RS) parameters are given in Table ??. A 50% carbon content is assumed, and the resulting residue is distributed across months following ?.

## 2.4 Soil-erosion factors

Soil-erosion factors estimate the annual rate of soil loss ($\mathrm{t\,ha^{-1}\,yr^{-1}}$) using the Revised Universal Soil Loss Equation (RUSLE):

$$RE = R \cdot K_{st} \cdot LS \cdot C \cdot P, \tag{5}$$

where $R$ is rainfall erosivity, $K_{st}$ soil erodibility corrected for the skeletal fraction, $LS$ the slope-length and steepness factor, $C$ the cover-management factor, and $P$ the support-practice factor. Because $R$, $K_{st}$, and $LS$ are independent of land use, a single $R \cdot K_{st} \cdot LS$ base raster is precomputed from the GloSEM layers (**??**) and reused for every commodity. Following common practice in LCA characterization, $P$ is set to 1. The cover-management factor is disaggregated into a crop component and a management component,

$$C_{\text{arable}} = C_{\text{crop}} \times C_{\text{mgmt}}, \qquad C_{\text{mgmt}} = C_{\text{till}} \times C_{\text{res}} \times C_{\text{cover}}, \tag{6}$$

where crop $C$ values are assigned per commodity from **?**, $C_{\text{till}} \in \{1, 0.35, 0.25\}$ for conventional, reduced, and no-till practices respectively, $C_{\text{res}} = 1 - 0.12\,F_{\text{res}}$ for the fraction of land leaving residues, and $C_{\text{cover}} = 1 - 0.2\,F_{\text{cover}}$ for the fraction using cover crops (**?**). The baseline factor assumes no erosion-reducing practices; other scenarios are obtained by multiplying the baseline by the corresponding $C_{\text{mgmt}}$.

## 2.5 Terrestrial-acidification factors

Terrestrial-acidification factors are obtained directly from the fate ($FF$) and soil sensitivity ($SF$) factors of **?**, supplied on a 2°×2.5° grid as the product $FF \times SF$ (in $\mathrm{mol\ H^+\,L^{-1}\,m^2\,kg^{-1}\,yr}$), with corrections to the original calculations consistent with IMPACT World+ v2.0 (**?**). No process modelling is performed; the published values are renormalized to $SO_2$-equivalents per kilogram of gas emitted. Per-cell gas emissions of $NH_3$, $NO_x$, and $SO_2$ are joined to the grid; the $SO_2$ acidification potential is computed as $FF \times SF$ times emissions; a normalization factor $NF_{SO_2} = AP_{SO_2}/E_{SO_2}$ is formed from the global $SO_2$ acidification potential and emissions; and each gas factor is obtained as $AP^{SO_2}_{\text{cell,gas}} = AP^{H^+}_{\text{cell,gas}}/NF_{SO_2}$, yielding factors in kg $SO_2$-eq per kg emitted. Factors are then harmonized to the common grid and aggregated; no outlier filtering is needed given their smooth distribution and coverage.

## 2.6 Software implementation

The full workflow is implemented in the open-source `sbtn_leaf` Python package. Core modules separate concerns: `RothC_Core` provides a single-point RothC implementation (useful for one farm or production unit), while `RothC_Raster` provides the GIS implementation that ingests multi-band GeoTIFFs and returns annual SOC snapshots; `cropcalcs` prepares crop-specific yields, residues, and plant-cover layers; `PET` implements the Thornthwaite and FAO-56 calculations; `map_calculations` performs resampling, raster arithmetic, and area-weighted zonal statistics; `map_plotting` produces the figures; and `data_loader` lazily loads and caches shared lookup tables. Thread-safe caches avoid redundant I/O for environmental and crop layers. Outputs are written as multi-band GeoTIFFs (native resolution: 1/12°, ≈9 km, for SOC; 25 km for erosion), as CSV summary tables, and as GeoPackages with a geometry layer and indicator layers for each reporting unit. The package ships with a unit-test suite covering pool initialization, rate modifiers, PET, yield-raster generation, resampling, zonal statistics, and output filtering, and is released under a CC0 license to maximize reuse.

### 2.7 Scenario design

Factors are computed for 42 land-use classes following **?**: 28 agricultural commodities, 15 forest types, and one grassland class. For cereals the model spans the combinations of two irrigation regimes (rainfed, irrigated), two tillage systems (conventional, reduced), and two residue-management options (residues left or removed), giving up to eight SOC scenarios per crop. Soil erosion uses four scenarios per crop, as it does not distinguish irrigation. This scenario grid lets companies evaluate not only their current land use but also the projected effect of switching practices.

## 3 Results

### 3.1 Global soil organic carbon factors

The published SOC LEAFs project SoilGrids 2016 baselines forward to 2030 under continued land use. Across terrestrial ecoregions and all crop scenarios, factors average 34.7 t C/ha (median 28.1), with the bulk of values between roughly 10 and 80 t C/ha and a long upper tail in cold and high-organic soils. Figure **??** shows example global SOC maps from the maize tutorial, in which conventional management drives substantial SOC depletion across much of the cropping area, consistent with the model's representation of residue inputs failing to offset decomposition under continuous cultivation. Figure **??** shows the multi-year SOC trajectory for a representative scenario, illustrating the gradual approach toward a new equilibrium.

### 3.2 Effect of land-management practices on SOC

Because the same baseline is run under multiple management combinations, the factors isolate the marginal effect of a practice change. Comparing conventional with reduced tillage for rainfed barley with residues retained, across the 4,599 ecoregions where both scenarios are defined, mean projected SOC rises from 31.4 to 33.0 t C/ha—a gain of about 1.6 t C/ha, or ≈5%. The benefit varies spatially with soil texture and climate, reflecting the sand- and SOC-dependent tillage factor adapted from **?**. These practice-resolved differences are the quantity most directly useful for target setting, since they estimate how far a management shift moves a landscape toward safe conditions.

### 3.3 Global soil-erosion factors

Baseline soil-erosion factors, aggregated to ecoregions across all land-use classes, average 48.8 t $\mathrm{ha^{-1}\,yr^{-1}}$ with a much lower median of 15.3 t $\mathrm{ha^{-1}\,yr^{-1}}$, reflecting a strongly right-skewed distribution in which steep, erosive, high-rainfall landscapes dominate the mean. Conservation practices reduce these rates multiplicatively: reduced tillage alone scales erosion by $C_{\text{till}} = 0.35$ and no-till by 0.25, with further reductions from residue retention and cover crops, so that the modelled erosion factor under a full conservation package can fall to a small fraction of the baseline.

### 3.4 Terrestrial-acidification factors

The acidification factors differ markedly among gases and biogeographic realms. Ecoregion mean factors are 0.24, 0.80, and 1.33 kg $SO_2$-eq per kg for $NO_x$, $SO_2$, and $NH_3$ respectively, with maxima exceeding 25 for $NH_3$ in the most sensitive soils. For $NO_x$, realm-mean factors range from 0.018 in Oceania and 0.061 in Australasia to 0.31 in the Nearctic and 0.48 in the Palearctic (Figure **??**), spanning more than an order of magnitude. This variability underscores why spatially explicit

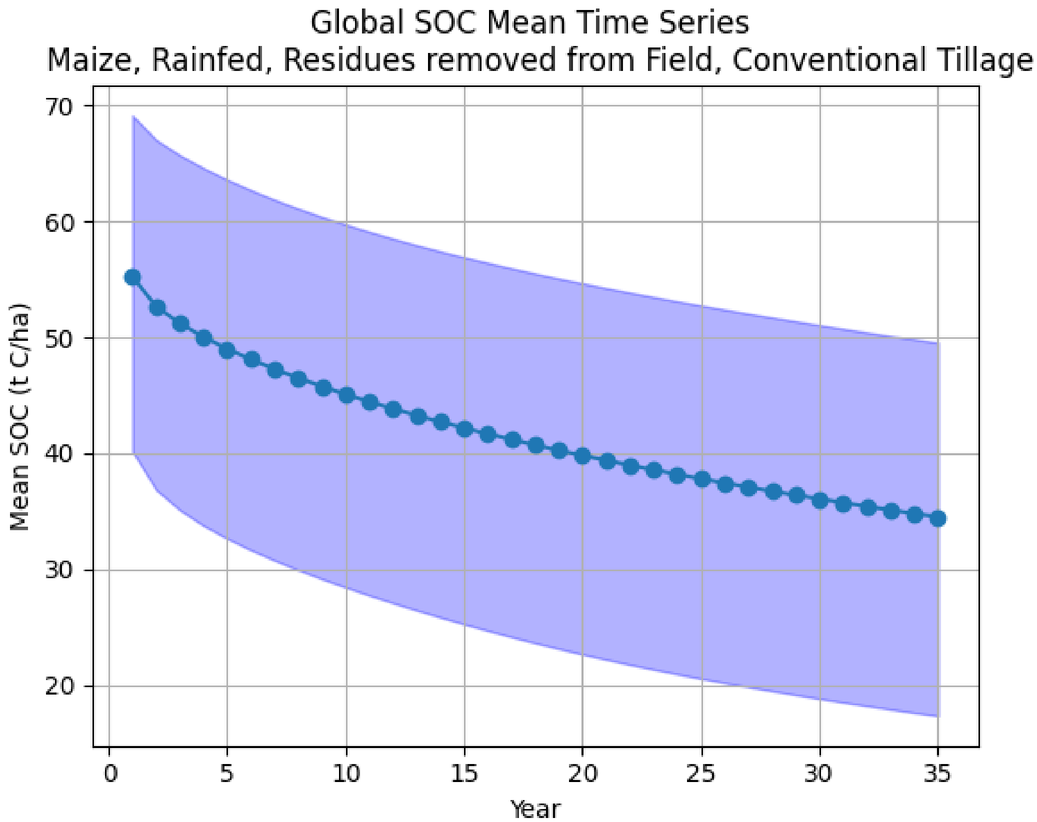


**Figure 2:** Example global SOC factor map (t C/ha) for a single crop and management scenario, generated by the `sbtn_leaf` pipeline. Cool-to-warm shading spans low to high SOC stocks; colour limits use distribution-based cutoffs to suppress extreme outliers.

factors matter: a single global value would misrepresent acidification impacts by realm by a large factor.

### 3.5 Validation of yield inputs

Because residue inputs depend on yields, the gap-filling procedure (Section **??**) was examined against its FAOSTAT and SPAM sources. Raw SPAM yields contain implausibly high values in a small number of pixels; restricting the comparison to the 5th–95th percentile range removes these artifacts and brings the spatial yield distribution into agreement with national statistics. This filtering, and the hierarchical fallback that prefers spatially disaggregated SPAM data and only resorts to ecoregion, biome, and nearest-neighbour estimates where necessary, limit the propagation of yield outliers into the SOC factors.

## 4 Discussion

**Significance.** LEAFs provide, to our knowledge, the first openly licensed, globally complete, ecoregion-resolved set of default factors spanning three soil-quality indicators and built specifically for science-based land target setting. By combining a process model (RothC) for SOC with empirical models for erosion (RUSLE) and acidification (fate–sensitivity factors), and by resolving

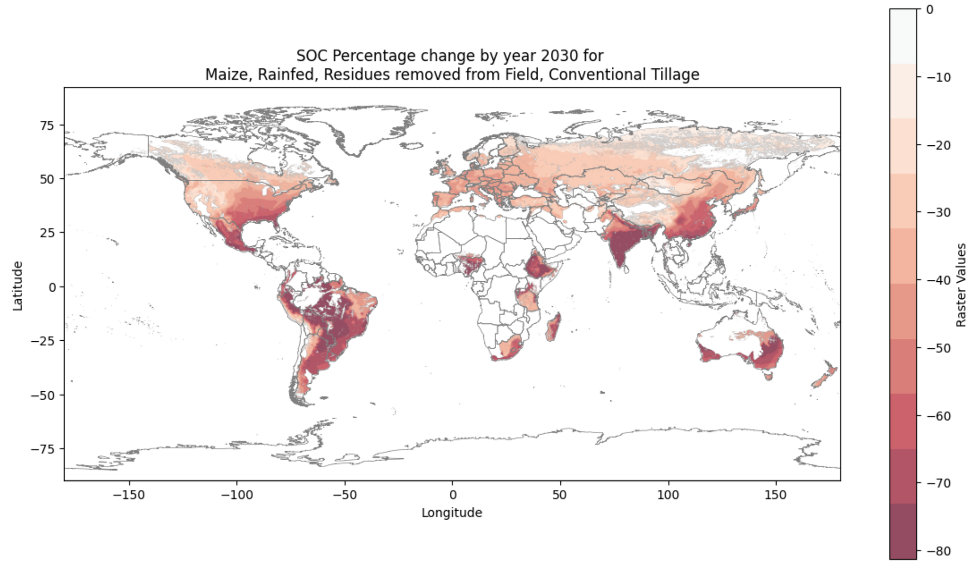


**Figure 3:** Projected SOC trajectory over the simulation horizon for a representative maize scenario, showing the modelled change in soil carbon as land use is held constant to 2030.

management scenarios explicitly, the dataset lets companies estimate both their current position relative to ecoregional thresholds and the expected effect of changing practices—all without primary field data, while retaining the option to substitute such data through the same reproducible pipeline.

**Relation to prior work.** The SOC component extends the global RothC implementation of **?** and **?** with practice-resolved scenarios, an updated conservation-tillage factor (**?**), and a transparent, tested software package. The erosion and acidification components adopt established global datasets (**???**) and harmonize them onto the same spatial framework, which is itself a contribution: a single, consistent grid and aggregation scheme makes the three indicators directly comparable across geographies.

**Limitations.** Several simplifications should be borne in mind. SOC factors use a 15 cm effective sampling depth as a proxy dictated by the 0–30 cm SoilGrids product, and a single 2016 baseline projected forward under constant land use; companies with land-use history can instead reconstruct current SOC and project from there using the documented procedure. The DPM/RPM partition is held at standard literature values rather than calibrated locally. Soil-erosion factors inherit the 25 km resolution and assumptions of the GloSEM base layers and adopt the common $P = 1$ simplification, which omits support practices such as terracing and contouring. Yield gap-filling, while guarded by percentile filtering, introduces uncertainty in data-sparse regions. Acidification factors are inherited from a published method and are not re-modelled. Finally, ecoregion and especially country averages mask within-unit heterogeneity; native-resolution rasters should be preferred where the geometry of a sourcing landscape is known.

**Future work.** Planned extensions include reporting and refining the effective UHTH resolution, adding farmyard-manure inputs and additional commodities to the SOC scenarios, propagating input uncertainty into factor uncertainty, and publishing a permanent archive of the native-resolution rasters alongside the summary tables.

*Figure placeholder — terrestrial-acidification factor ($SO_2$-eq) by biogeographic realm, shown as boxplots per realm for $NO_x$, $SO_2$, and $NH_3$. Regenerate by running `paper/Paper_Graphs_Figures.ipynb` (section "Terrestrial Acidification").*

**Figure 4:** Distribution of terrestrial-acidification factors across ecoregions, grouped by biogeographic realm. Realm means for $NO_x$ range from ≈0.02 (Oceania) to ≈0.48 (Palearctic) kg $SO_2$-eq per kg emitted.

# 5 Data and Code Availability

The complete source code (the `sbtn_leaf` package), theoretical documentation, executable example notebooks, and precomputed factor tables for SOC, soil erosion, and terrestrial acidification are available in the project repository under a CC0 1.0 Universal (public-domain) dedication. Summary factors are provided as CSV tables and GeoPackages for ecoregion, country, and subcountry units; native-resolution rasters are distributed separately because of their size. The example notebooks reproduce the full workflow from harmonized inputs to exported factors, enabling users to regenerate the published factors or to derive customized factors from primary data.


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